ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

AGRICULTURAL SCIENCES

PAPER 1

MARKS: 100 **TIME: 2 hours**

Topics: soil science, agricultural ecology, and plant studies (photosynthesis, respiration and plant nutrition).

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Number your answers correctly.
2. Write in full sentences where explanations are asked for. You may use a non-programmable calculator.

SECTION A

QUESTION 1

1.1 Choose the correct answer. Write only the letter (A–D).

1.1.1 The soil particle with the smallest size is …

A sand. B silt.
C clay. D gravel.

1.1.2 A soil with a pH of 5 is …

A acidic. B neutral.
C alkaline. D saline.

1.1.3 The process in which plants make glucose using light energy is …

A respiration. B photosynthesis.
C transpiration. D osmosis.

1.1.4 Which nutrient is needed in the LARGEST amounts by plants and promotes leaf growth?

A Nitrogen B Iron
C Zinc D Copper

1.1.5 Organisms that make their own food are called …

A consumers. B decomposers.
C producers. D parasites.

(5 × 2)
(10)

1.2 Give ONE term for each description.

1.2.1 The breaking down of rocks into smaller particles by physical, chemical or biological processes

1.2.2 Dark, decomposed organic material in the soil

1.2.3 The relative proportions of sand, silt and clay in a soil

1.2.4 A large area with a typical climate, vegetation and animals, such as the savanna or grassland

1.2.5 The loss of water vapour from the leaves of a plant (5 × 2)
(10)

1.3 Change the UNDERLINED word(s) to make each statement TRUE.

1.3.1 <u>Sandy</u> soils hold the most water.

1.3.2 Agricultural lime is added to soil to lower the pH of alkaline soils.

1.3.3 During respiration, plants release oxygen and take in glucose and oxygen.

1.3.4 Earthworms are the main decomposers of organic matter that are too small to see without a microscope.

1.3.5 A shortage of phosphorus causes yellowing of the older leaves (chlorosis) in maize. (5 × 2) (10)

TOTAL SECTION A: 30

SECTION B

QUESTION 2: SOIL SCIENCE

2.1 Name the FIVE factors that influence soil formation. (5)

2.2 A farmer tested three soil samples. The results are shown below.

Sample	Sand %	Silt %	Clay %	pH
A	80	10	10	6,5
B	40	40	20	6,8
C	15	25	60	5,2

2.2.1 Which sample is most likely a loam soil? Give a reason. (2)

2.2.2 Which sample will drain water most quickly? Explain. (3)

2.2.3 Which sample needs lime? Give a reason. (2)

2.2.4 Give TWO problems a farmer may have with a soil like sample C. (2)

2.3 Give FOUR ways in which organic matter improves soil. (4)

2.4 Name THREE methods a farmer can use to prevent soil erosion. (3)

2.5 Explain how soil compaction harms plant growth. (4)

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QUESTION 3: AGRICULTURAL ECOLOGY

3.1 Study the food chain: Grass → Cattle → Humans

3.1.1 Identify the producer and the primary consumer. (2)

3.1.2 Explain why only about 10% of the energy is passed on from one level to the next. (3)

3.2 Distinguish between biotic and abiotic factors in an ecosystem. Give TWO examples of each. (6)

3.3 Name TWO biomes in South Africa that are suitable for cattle farming, and give a reason for ONE of them. (3)

3.4 Explain how overgrazing damages the veld, and suggest TWO ways to prevent it. (6)

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QUESTION 4: PLANT STUDIES

4.1 Write the word equation for photosynthesis. (3)

4.2 Name THREE factors that affect the rate of photosynthesis. (3)

4.3 Farmers often grow vegetables in tunnels (greenhouses). Explain TWO ways in which this increases the rate of photosynthesis. (4)

4.4 Tabulate TWO differences between photosynthesis and respiration. (5)

4.5 A fertiliser bag is labelled 2 : 3 : 2 (22).

4.5.1 Name the THREE nutrients in the order shown on the label. (3)

4.5.2 What does the number (22) mean? (2)

4.5.3 Calculate the mass of nitrogen in a 50 kg bag of this fertiliser. Show your working. (3)

4.6 Give ONE advantage and ONE disadvantage of using kraal manure instead of chemical fertiliser. (2)

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TOTAL SECTION B: 70 · GRAND TOTAL: 100

END OF QUESTION PAPER